Limitations of Existing Studies

Many studies report gains, but few compare methods fairly.

Study A	Study B	Study C	Study D	...
A > baseline	B > baseline	C > baseline	D > baseline	
2B ALFWorld	4B WebShop	9B ALFWorld	2B WebShop	
Custom train	Different tuning	Larger budget	Different eval	

Baseline scores reused across papers

Prior baseline scores

Reported results



Which gains can we rely on?

Fair comparisons require costly reproduction.

Key evaluation evidence is often missing

1/10	Repeated evaluation	**5/10**	Training failure analysis	**3/10**	Measured cost

Evaluate together

A Unified Benchmark

Self-Distillation in Agentic RL

11 methods

GRPO baseline

10 representative methods

3 model scales

2B 4B 9B

4 benchmarks

ALFWorld

ScienceWorld

Search-QA

WebShop

132	training runs	**12,124.4**	H800 GPU-hours

Key findings

- **Gains are setting-dependent**
- **Final scores hide instability**
- **Higher cost, not always better**